

5 (**short**) An incompressible flow of air with density ρ travelling at velocity V_1 through a circular, straight pipe of diameter D_1 with pressure p_1 undergoes a smooth contraction to diameter D_2 . Two Pitot tubes connected by a manometer with closed valves are installed upstream and downstream of the contraction, as indicated in Fig. 3. Friction forces may be neglected throughout.

- (a) Find the velocity downstream of the contraction, V_2 , in terms of V_1 , D_1 and D_2 . [2]
- (b) Find the pressure downstream of the contraction, p_2 , in terms of p_1 , ρ , V_1 , D_1 and D_2 . [4]
- (c) If the valves shown in the figure are opened, describe what happens to the height difference across the manometer, Δh , and justify your answer. [4]

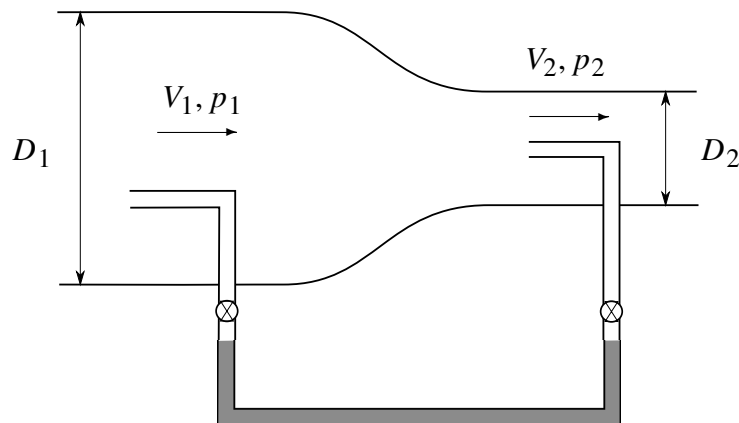


Fig. 3